Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/27/2013
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Manipulation of two-phase flow in low-gravity environments: drop-phase relocation by active wettability control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ithaca	**State:**	NY
Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:			
Flight Assignment:	NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012) NOTE: extension to 5/31/2013 is in process, per PI (Ed., 3/6/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Daniel, Susan (Cornell University) Hui, Chung Yuen (Cornell University)		
Grant/Contract No.:	NNX09AI83G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The broad technological goal of this project is to control two-phase flow in low gravity with active wettability. The scientific goal is to move droplets in a directed fashion using passive and active surface-energy gradients. The latter would yield active-wettability control. Fundamental to both goals is the stability of the sessile drop, both natural and forced response. Accordingly, sub-tasks are identified: I: Sessile-drop stability; and II: Active wettability-control. To electronically control wettability of a substrate surface in real-time would enable manipulation of two-phase flow in the absence of Earth's gravity. Condensers and separators are a crucial part of life support and power systems. A proof-of-concept device for collecting drops on a surface in low-gravity is the goal. Motivation comes from earth-based observations that passive chemical gradients can propel condensing steam droplets to a collector with centimeter-per-second speeds. Note that thermal effects are not central to the mechanism. The direction of movement of a drop is biased by the chemical gradient, and the energy to propel the droplet is provided by plane-normal oscillations. The oscillations also provide fluctuations sufficient to de-pin the contact-line. It is proposed to develop a tunable gradient to serve as a controllable bias and to develop a local vibration source to create the fluctuations. The goal is the capability to move drops on-demand in any number of directions. A scientific basis for the technology requires understanding the nature of the contact-line, contact-angle and spreading behavior. Predictions of which drop oscillations will be most effective in de-pinning the contact-line and which vibrations will be most effective in		

	propelling the drops are being carried out. Experiments to test the predictions are being conducted.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The location of the liquid- and gas-phases in two-phase liquid/gas flows on Earth depends crucially on the strength of Earth's gravity. Many devices that perform well on Earth are unreliable for lunar and Mars missions. They falter in low-gravity. For example, the Total Organic Carbon [1] analyzer that works well on Earth is 'currently not in use due to difficulty in bubble removal.' Hence, monitoring the quality of on-board water supplies becomes a high risk (red-light) issue for lunar and Mars missions (risk 38 of the Bioastronautic Critical Path Roadmap (BCPR) [1]). This is not an isolated instance. Condensers are crucial to Rankine-cycle power plants [2], among other applications, and for regenerative Life Support systems (CO2 condenser/separators) [3]. Additionally, success of this research effort will likely lead to better Earth-based two-phase flow systems, including condensers and other heat-exchange devices. 1. Center, J. S., Bioastronautic Critical Path Roadmap, Supporting Documents. 2. Lahey, R. T.; Dhir, V., Research in Support of the Use of Rankine Cycle Energy Conversion Systems for Space Power and Propulsion. NASA/CR-2004-213142 2004. 3. [NAP2000], Microgravity Research in Support of Technologies for the Human Exploration and Development of Space and Planetary Bodies. National Academy Press: 2000.
Task Progress:	1) The shapes and frequencies of a drop held by surface tension and undergoing inviscid motions, sitting on a solid substrate subject to pinned and mobile contact-line conditions, has been solved. Predictions of the spectra and shapes as they depend on contact-angle and mobility parameter have been tested in experiment. This stability of the sessile drop is a long-standing problem of great relevance to space applications. Our seminal results will also benefit Earth-based applications from spray atomization to micro-printing to microfluidic device design. 2) Deformable drops coupled together are subject to volume-scavenging instabilities. In the regime of inviscid motions, coupled droplets can exhibit periodic, quasi-periodic, or chaotic behavior. For viscous motions, the system typically drives toward an equilibrium characterized by having the least interfacial area, consistent with fixed volume and constraint. 3) Active-wettability control using surface energy gradients has been demonstrated.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
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